Evaluating expressions with and without changing the subject

1 Substitute means to put in place of another. In algebra, substitution can be used to replace ___________ with values. Tick **1** correct answer

- [ ] constants
- [ ] equations
- [ ] solutions
- [x] variables

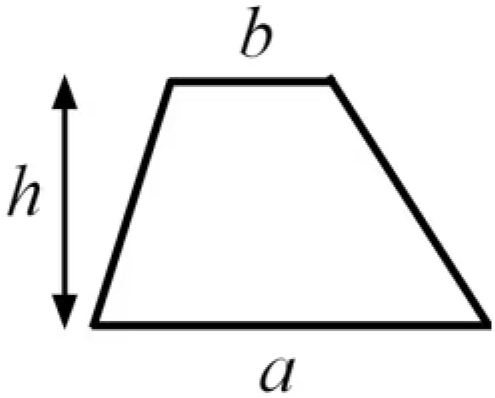
2 $A = \frac{1}{2}(a + b)h$ is the formula for area of a trapezium. It can be rearranged to $\frac{2A}{h} - a = b$ in order to make b the *subject* of the formula. Fill in the blank

3 The formula for area of a triangle is $A = \frac{1}{2}bh$. The area of a triangle with base 7.5 cm and height 6 cm is *22.5* cm² Fill in the blank

4 The formula $s = \frac{d}{t}$ can be rearranged to $t = \frac{d}{s}$ or $d = s \times t$. If $d = 12$ and $s = 3$, find t. Tick **1** correct answer

- [ ] 36
- [x] 4
- [ ] $\frac{1}{4}$

5 Find the area of a trapezium with $h = 7, a = 3.2, b = 4.8$. Tick **1** correct answer



$$A = \frac{1}{2}(a+b)h$$

- [ ] $A = \frac{1}{2}(3.2 + 4.8) + 7 = 11$ square units
- [x] ✓ $A = \frac{1}{2}(3.2 + 4.8) \times 7 = 28$ square units
- [ ] $A = 2(3.2 + 4.8) \times 7 = 112$ square units
- [ ] $A = \frac{1}{2}(3.2 \times 4.8) \times 7 = 53.76$ square units

6 $A = \frac{1}{2}(a + b)h$ is the formula for area of a trapezium. Select all the correct rearrangements of this formula. Tick **2** correct answers

- [x] ✓ $\frac{2A}{h} - b = a$
- [ ] $\frac{Ah}{2} - a = b$
- [ ] $\frac{A}{2h} - a = b$
- [ ] $\frac{2h}{A} - a = b$
- [x] ✓ $\frac{2A}{a + b} = h$